

Kerala Polytechnic Syllabus

Electronic Instrumentation Lab

Course Code: 2089

Semester: 2

Revision: 2021

[View Full Course Content Below](#)

Course Overview

Program: Diploma in Instrumentation Engineering

Category: Engineering Science

Credits: No Credit (Lab Course)

Total Periods: 45 Hours

Assessment: CIE (Continuous Internal Evaluation) & ESE (End Semester Exam)

Objective: This lab introduces students to the practical aspects of measuring instruments, including bridge circuits, calibration of meters, and the use of modern diagnostic tools like DSO and Strip Chart Recorders.

Course Outcomes (CO)

CO No.	Description	Cognitive Level
CO1	Determine static characteristics of instruments and convert galvanometers to ammeters/voltmeters.	Applying
CO2	Determine unknown R, L, C values using appropriate bridges (Wheatstone, Hay's, Schering).	Applying
CO3	Calibrate Wattmeter and Energy meter using direct loading.	Applying
CO4	Explain and utilize CRO/DSO and recorders for signal measurement.	Understanding

Module 1: Static Characteristics & Galvanometer Conversion

12 Hours

▼ M1.01: Static Characteristics & Calibration

Static characteristics are the properties of an instrument when the input is either constant or varying very slowly.

- **Accuracy:** The degree of closeness to the true value.
- **Precision:** The degree of repeatability of measurements.
- **Error:** The difference between measured value and true value.
($\text{Error} = A_m - A_t$)
- **Sensitivity:** The ratio of change in output to the change in input.
- **Hysteresis:** The difference in readings when a value is approached from an increasing direction versus a decreasing direction.

Malayalam support:

Instrument (Instrument) 精度 (Accuracy) 和 准确度 (Precision) 是 仪器 的重要 性能 指标。 精度 是指 仪器 测量 结果 与 真值 的 接近 程度。 准确度 是指 仪器 测量 结果 的 分散 程度。 精度 和 准确度 是 仪器 设计 和 使用 中 必须 考虑 的 重要 因素。

▼ M1.02: Galvanometer Conversion

A D'Arsonval galvanometer can be converted into an Ammeter or Voltmeter by adding specific resistances.

1. Conversion to Ammeter

A low resistance called **Shunt (\$R_s\$)** is connected in parallel with the galvanometer.

$$R_s = R_g / (n - 1)$$

Where $n = I / I_g$ (Multiplying factor), I is the total current to be measured, and I_g is full-scale deflection current.

2. Conversion to Voltmeter

A high resistance called **Multiplier (\$R_m\$)** is connected in series with the galvanometer.

$$R_m = R_g (n - 1)$$

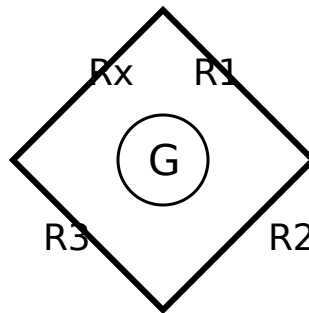
Where $n = V / V_g$ (Multiplying factor), V is the total voltage to be measured.

Module 2: Measurement using Bridges

8 Hours

▼ M2.01: Wheatstone's Bridge (Resistance)

Used for measuring medium resistances (1Ω to $100k\Omega$). It consists of four arms, a DC source, and a galvanometer.



Wheatstone Bridge Circuit Diagram

Balance Condition: $R_x = (R_2 / R_1) \times R_3$

▼ M2.02: Hay's Bridge (Inductance)

Specifically used for measuring high-Q inductors ($Q > 10$). It differs from Maxwell's bridge as the capacitor and resistor are in series.

Malayalam support:

High Quality factor (Q) is a measure of the sharpness of resonance in a series R-L-C circuit. Hay's Bridge is used for measuring the quality factor of a coil.

▼ M2.03: Schering Bridge (Capacitance)

Used for measuring unknown capacitance, dielectric loss, and power factor. It is one of the most important AC bridges.

Module 3: Calibration of Wattmeter & Energy Meter

8 Hours

Direct Loading Method

In this method, the meter is connected to a real load (like a lamp bank or rheostat). The reading of the meter under test is compared with a standard (calibrated) meter.

Percentage Error Calculation:

$$\% \text{ Error} = \left[\frac{(\text{Measured Value} - \text{True Value})}{\text{True Value}} \right] \times 100$$

Energy Meter Calibration:

Energy meter 0000000000 0000000000 000000 (Revolutions) 0000000000
0000000000 000000000000 000000000000. 00000000 000000000000 00000000
000000000000 0000 000000000000.

Module 4: Oscilloscopes & Recorders

14 Hours

▼ M4.01 - M4.03: CRO and DSO Applications

CRO (Cathode Ray Oscilloscope) and **DSO (Digital Storage Oscilloscope)** are used to visualize waveforms.

- **Voltage Measurement:** $V_{p-p} = (\text{Vertical divisions}) \times (\text{Volts/div setting})$
- **Time Period:** $T = (\text{Horizontal divisions for 1 cycle}) \times (\text{Time/div setting})$
- **Frequency:** $f = 1/T$

Lissajous Patterns

Used for frequency and phase measurement by applying two different signals to the X and Y plates.

Pattern Shape	Phase Difference (ϕ)
Straight Line (Diagonal)	0° or 360°
Circle	90° or 270°
Ellipse	Between 0° and 90°

▼ M4.04: Strip Chart Recorder

A device that records a continuous visual record of a signal on a moving strip of paper.

- **Drive Mechanism:** Moves the paper at a constant speed.
- **Stylus/Pen:** Moves across the paper proportional to the input signal.

Formula Bank

Ammeter Shunt

$$R_s = R_g / (I/I_g - 1)$$

Voltmeter Multiplier

$$R_m = R_g * (V/V_g - 1)$$

Wheatstone Balance

$$R_x = (R_2 * R_3) / R_1$$

Frequency (CRO)

$$f = 1 / (\text{Div} \times \text{Time/Div})$$

Ammeter Shunt Calculator

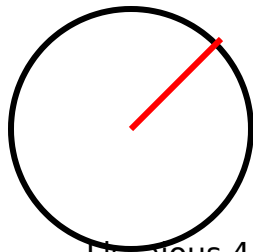
Galvanometer Resistance (R_g) Ω :

Full Scale Current (I_g) mA:

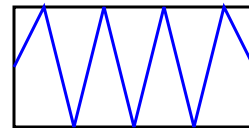
Desired Range (I) A:

Calculate Shunt (R_s)

Visual Library



Phase Measurement



Strip Chart Trace

Practice Questions

1. Define Hysteresis in instruments.

Hysteresis is the maximum difference in output readings for the same input, when the input is approached from opposite directions (increasing and decreasing).

2. Why is a shunt used in an ammeter?

A shunt is used to bypass the majority of the current around the galvanometer coil, allowing the meter to measure currents much larger than its full-scale deflection current without damaging the coil.

3. Which bridge is best for measuring high-Q inductance?

Hay's Bridge is preferred for high-Q inductors ($Q > 10$).

Practice Model Paper

END SEMESTER EXAMINATION - PRACTICE

Course: Electronic Instrumentation Lab (2089)

Part A: Short Answer (Select any 2)

1. Explain the difference between Accuracy and Precision.
2. State the balance condition for a Wheatstone bridge.
3. What is a Lissajous pattern?

Part B: Practical/Long Answer

1. Describe the procedure to convert a 1mA, 100 Ω galvanometer into a 0-10A ammeter. Calculate the shunt resistance.
2. Explain the calibration of a single-phase energy meter using direct loading method with a neat circuit diagram.

Quick Revision

- **Static Characteristics:** Accuracy, Precision, Sensitivity, Hysteresis, Resolution.

- **Ammeter:** Low resistance shunt in parallel.
- **Voltmeter:** High resistance multiplier in series.
- **Wheatstone Bridge:** Measures medium resistance.
- **Schering Bridge:** Measures capacitance.
- **Hay's Bridge:** Measures high-Q inductance.
- **CRO:** Measures Voltage, Time, Frequency, and Phase.

Glossary

DSO

Digital Storage Oscilloscope - An oscilloscope that digitizes and stores the input signal.

Loading Effect

The distortion of the signal being measured caused by the act of connecting the measuring instrument.

Q-Factor

Quality factor; a dimensionless parameter that describes how underdamped an oscillator or resonator is.

Official Source Declaration

This content is based on the official **Revision 2021** syllabus provided by the **State Institute of Technical Teachers' Training & Research (SITTTR)**, Kerala, for the course **Electronic Instrumentation Lab (Course Code: 2089)**.

Source URL: [SITTTR Syllabus Website](#)

മലയാളം പിന്തുണ (Malayalam Support)

പ്രധാന പദങ്ങൾ:

- **Bridges:**

വ്യാപ്തം, വോൾട്ടേജ്, റെസിസ്റ്റൻസ് തുടങ്ങിയവയെ അളക്കാനുള്ള ഉപകരണങ്ങൾ.

- **Calibration:**

□□□□□□ □□□ □□□□□□□□□□ □□□□□□□□□□ □□□□□□ □□□□□□□□□□.

□□□□□□□□□□. □□□□□□□□□□, □□□□□□□□□□ □□□□□□ □□□□□□□□□□□□□□□□.